The Human Auditory System

SYNOPSIS

Sound arrives at you as a pressure variation in air, and what reaches your brain is a nerve signal. Something has to do that conversion, and it is the ear. This note follows one sound through it, from the outside of your head to the nerve, and asks what each part is for.

PERFORMANCE EXPECTATIONS

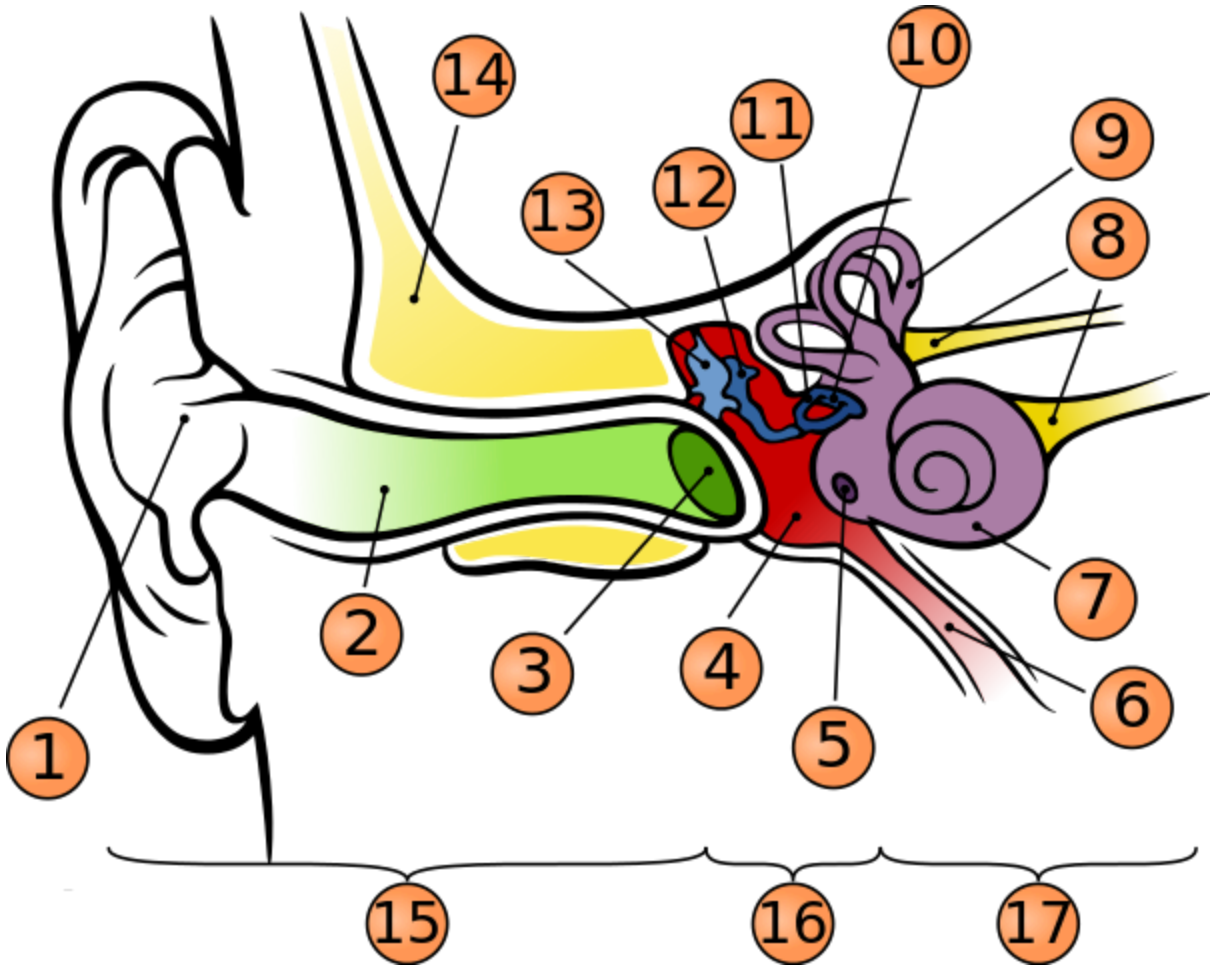
Identify the major components of the auditory system and say which of the three divisions each one belongs to. Describe what each division does to the sound: the outer ear collecting it, the middle ear passing it from air into fluid, the inner ear turning it into nerve impulses. Explain why a boundary between air and fluid needs matching at all, and what the middle ear does about it. Describe how loud sound causes hearing loss, and ways to protect your hearing.

Instructions and Questions

Module 2 ended with sound as a pressure wave travelling through air. This note is about what happens when that wave runs into the side of your head. Nothing in here changes the sound — the ear does not make it louder or alter its frequency. The ear's whole job is **conversion**: pressure in air at one end, electrical impulses on a nerve at the other.

Three parts, one chain

The ear is usually divided into three, and the divisions are not arbitrary — each one works in a different medium. The **outer ear** is open to the air. The **middle ear** is an air-filled cavity behind the eardrum. The **inner ear** is filled with fluid. Sound has to cross all three.



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Figure 1. Write the name of each numbered part on its line. 15, 16 and 17 are the three divisions. Then trace the path one sound takes, left to right, with your finger.

Figure after Chittka L, Brockmann, *Anatomy of the Human Ear*, CC BY 2.5, via Wikimedia Commons; numbered version derived by Sgbeer. Same figure used in the PHY 105 reference deck.

1 What are the three major parts of the ear, and what is the one thing that makes them three different parts rather than one?

The outer ear: collecting

The **auricle** is the fleshy part you can see. The **ear canal** — the external auditory canal — is the tube running from the outside world in to the **tympanic membrane**, which is the eardrum. In your own words from last year: the outer ear *collects sound and funnels it into the inner parts*. That is the whole of its job.

2 You cup a hand behind your ear to hear someone better across a room, and it works. What have you actually changed? Be careful: you have not made the talker any louder.

> **LOOKING AFTER THE CANAL**
>
> The canal is a dead end sealed by a membrane you cannot replace, and nothing rigid should go down it. If it needs cleaning, that is drops and a rubber bulb — not a cotton bud, which packs wax against the eardrum, and not anything you could light on fire.

The middle ear: passing it on

Behind the eardrum is the **tympanic cavity**, and bridging it are the three smallest bones in your body: the **hammer, anvil and stirrup** — the malleus, incus and stapes. They are hinged to each other so that they work as a **lever**. One end rests on the eardrum; the far end, the footplate of the stirrup, rests on the **oval window**, which is the door into the inner ear.

So the conversion in this division is **sound wave → mechanical vibration**. Air pushes the eardrum, the eardrum pushes the bones, the bones push the oval window.

One more opening matters here. The **Eustachian tube** runs from the tympanic cavity down to the back of the throat, and it exists to **equalise the pressure** on the two sides of the eardrum.

3 An aeroplane climbs and your ears block up. Swallowing or yawning fixes it. Using the Eustachian tube, say what went wrong and what swallowing did. Then say why your hearing was worse while it lasted.

Why the middle ear has to be there at all

It is worth asking why the eardrum does not simply sit on the fluid. Why is there a cavity with a lever in it at all? The answer is the one idea in this note that is pure physics, and it is called **impedance**.

A medium's **acoustic impedance** is how hard it is to get it moving — roughly, its density times the speed of sound in it. Air is light and easy to move. Water is about 3500 times harder. When a sound wave meets a boundary between two media whose impedances are very different, almost all of it **reflects** rather than crossing. At a plain air-to-water boundary, around **99.9%** of the energy bounces straight back.

The inner ear is full of fluid. So a sound going straight from air into it would almost entirely bounce off, and you would be very nearly deaf. The middle ear is the fix. It is an **impedance matcher**, and it works two ways at once.

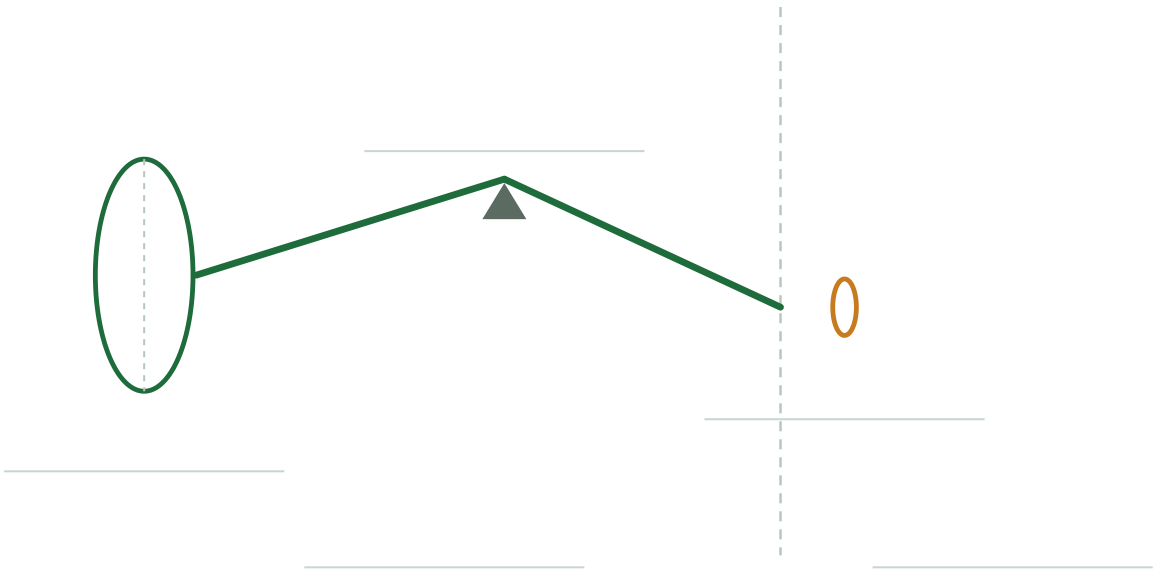


Figure 2. The eardrum and the oval window are drawn to their true relative size. Label each, and label what is on each side of the dashed line. Then write on the drawing what happens to the force and to the area between the one and the other.

4 The eardrum has about 17 times the area of the oval window. Take the same total force arriving at each. What does that do to the **pressure**, and why is pressure the thing that matters here rather than force?

5 Someone's ossicles have fused and can no longer move. Sound still reaches the eardrum and the inner ear is perfectly healthy. Can they hear? Explain using the figure.

The inner ear: making the signal

The **cochlea** is a fluid-filled, snail-shaped spiral — your own description of it. The stirrup drives the **oval window** at one end; the **round window**, just below it, bulges out as the oval window pushes in. Running the length of the spiral is the **basilar membrane**, and sitting on that is the **organ of Corti**.

The organ of Corti carries **hair cells**, named for the **stereocilia** standing up from them. When the fluid moves, the membrane above them shears across those stereocilia and bends them, and a bent hair cell fires. That is the moment the signal stops being mechanical: **mechanical vibration → electrical nerve impulse**, out along the cochlear nerve to the brain.

6 The fluid in the cochlea cannot be squashed. Given that, say why there has to be a round window, and what would happen without one.

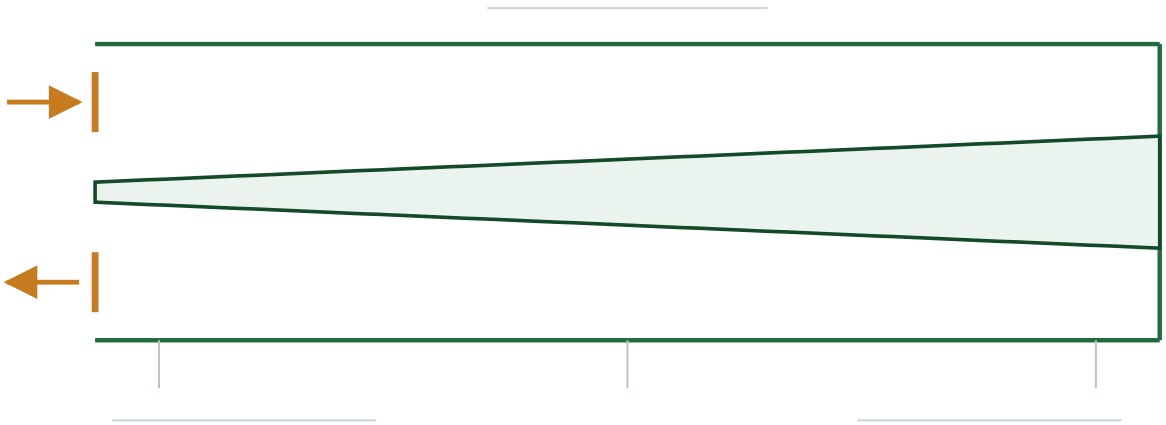


Figure 3. The membrane's shape is drawn for you. Mark the base and the apex, and write at each end what kind of frequency that end responds to. Mark where the oval window and the round window are.

7 Where along the membrane does a high-pitched whistle do most of its work — near the base or near the apex? And what does the brain actually use to tell one pitch from another?

8 The semicircular canals sit right next to the cochlea, are filled with the same kind of fluid, and send signals along a nerve of their own. They have nothing to do with hearing. What are they for, and why is it reasonable that they ended up here?

Hearing loss, and keeping yours

Everything above the hair cells is replaceable in principle — eardrums can be patched, ossicles can be replaced with prosthetic ones. The hair cells themselves are the problem: **you are born with all of them you will ever have, and they do not grow back.**

Loud sound damages them mechanically — the stereocilia are bent too far, too often, and break. Two things decide the damage: **how loud** and **for how long**. A short, very loud event and a long, moderately loud one can do the same harm, which is why a day at work in a noisy room is as real a risk as a concert. The loss shows up first in the **high** frequencies, because the base of the cochlea takes the brunt of every sound going in.

9 Noise-induced hearing loss usually appears first at high frequencies, around 4000 Hz, whatever the noise was. Using Figure 3, suggest why the base of the cochlea should suffer first.

10 Name three different ways to cut the dose your hair cells take. You should be able to get all three straight out of the two things that decide the damage.

11 A cochlear implant does not make sound louder. It runs a wire of electrodes into the cochlea and stimulates the nerve directly. From the chain in Figure 1, say which parts it is replacing and which part it absolutely depends on still working.

WHERE THIS IS GOING

Every question in this note has been about *how* the ear works, not *how much* it takes to hear something. Note 2 puts a number on that. Your ear covers a range of intensities so enormous that ordinary numbers become useless for it, which is where the decibel comes from.

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